

MECHANICS OF SARTORIUS MUSCLES IN UROMASTIX: AN EFFECT OF IMMOBILIZATION AND REMOBILIZATION

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ABSTRACT

Left limb of Uromastix casted to immobilize Sartorius (Sar) in stretched position. Right limbs being control (Con). After 30 days cast was removed and animals used in 2 groups. 1st for the isolation of Con and immobilized (Imb) Sar, used for mechanical recordings i.e., twitch (Tw) and tetanus (Tet). 2nd group was allowed to remobilize (10 days) and used for experiments at the end of remobilization (Rmb). Results of various parameters measured from Tw and Tet demonstrated significant fall in Tw and Tet tensions and their ratio after 30 days of Imb. Tw contraction time (TwCT) also reduced significantly after Imb. Whereas, Tw peak and active state durations reduced insignificantly. However, after Rmb the tension parameters recovered mostly but TwCT remained significantly less from its Con counterpart. It is concluded that, mechanical activity of Sar reduced by Imb represented an alteration in contractile to non contractile protein ratio and lesser ATPase activity. These changes are reversible on Rmb. The histological and biochemical studies are suggested to strengthen this conclusion.

Note: This study is a part of NSRDB research project leading to M.Phil. degree conducted at the Department of Physiology, University of Karachi.

INTRODUCTION

Immobilization of joints by plaster casting cause marked changes in the morphological, mechanical and biochemical characteristics of skeletal muscles (Booth and Kelso, 1973; Melichna and Gutmann, 1974; Lindboe and Presthus, 1985 and Kannus *et al.*, 1992). Various type of skeletal muscles are reported to show different effects of immobilization. Edgerton *et al.*, (1975) suggested that degree of atrophy and changes in muscle fibre composition that occur with prolonged immobilization are dependent on both level of activity of the muscle prior to immobilization and the degree of disturbance of its normal function during immobilization

Experiments have also been performed by a number of workers for the determination of recovery in the changes that were produced as a result of immobilization. However, the time course of recovery has been found to be greatly affected by the duration of immobilization (Witzmann *et al.*, 1982)

and type of the muscle (Booth and Seider, 1979). Finkelstein *et al.*, (1993) studied the recovery of muscle after different periods of denervation. They also reported that the force produced by reinnervated muscle depends on the period that the muscle was denervated. The experiments performed by Booth and Seider (1979) demonstrated faster recovery in slow-twitch muscle as compared to the fast-twitch muscles, also confirmed by Witzmann *et al.*, (1982). Booth and Seider (1979) reported faster recovery in muscular size and protein contents than the maximum isometric tension capacity. Further, Witzmann *et al.*, (1982) showed that tetanic tension required 28 days to recover fully whereas, the various time parameters recovered much earlier.

Recently, Cotter *et al.*, (1991) reported that natural recovery (2 weeks) had little effect. A prolongation of recovery period to 4 weeks increased the muscle fibre area but most functional characteristics remained impaired. Maxwell and Enwemeka (1992) also performed experiments regarding recovery and demonstrated that subsequent lengthening of muscles for 2 or 4 weeks didn't reverse the atrophy.

Most of the studies mentioned above were performed on mammalian muscles and almost negligible data is available regarding the effect of immobilization on skeletal muscles of reptiles. Recently, it has been suggested that skeletal muscles of reptile, *Uromastix hardwickii* behave differently to immobilization as compared to mammalian muscles. Azeem *et al.*, (1993) reported that immobilization for 7 and 20 days was unable to produce significant atrophy in the Gastrocnemius muscles of *Uromastix hardwickii*. They further reported significant decline in various contractile parameters and suggested that this decline was due to a probable change in energy producing substrates and a change in the sarcoplasmic capacity for calcium manipulation in immobilized muscles of *Uromastix hardwickii*.

It is interesting that immobilization of *Uromastix*'s muscle (Azeem *et al.*, 1993) didn't produce atrophy but reduces the contraction parameters significantly. However, it is not known that this reduction in contraction parameters is reversible on recovery or not. The present work was therefore designed to investigate the effect of immobilization and remobilization (by the removal of plaster casts) on Sartorius muscles of *Uromastix hardwickii*.

MATERIALS AND METHODS

The animals *Uromastix hardwickii* of both sexes were obtained (200-300 gms of body weight) and their left legs were immobilized in stretched knee position to immobilize Sartorius in neutral position. The right leg served as control. For the purpose of immobilization the animals were anaesthetized

by injecting xylocaine (0.05 gm/kg) intramuscularly (Africa *et al.*, 1993). After anaesthetization, left leg was stretched at knee joint and a thin layer of cotton was wrapped on it (to prevent occlusion of blood vessels). A strip (25 x 2.5 cm) of Gypsona plaster (Smith Kline & French Lab. Pakistan) was cut and applied over the leg to immobilize the knee joint in stretched position. After 30 days of immobilization, plaster castings were removed and animals were randomly divided into two groups.

The animals of 1st group decapitated for the isolation of Sartorius muscles from both the legs (i.e., right-control, left-immobilized). Muscles were stimulated supramaximally (current strength = 50v; duration = 5mS) to record twitch (1pulse/sec) and tetanus (80 pulses/sec). From these recordings tension and time parameters were measured and calculated (Azeem and Shaikh, 1987).

The animals of 2nd group allowed to remobilize the cast removal for 10 days and at the end of this animals were used for the recording of twitch and tetanus. Statistical analysis of the data was done by using LOTUS 123 RELEASE 4. For comparison purpose t-values were calculated to determine the significance levels.

RESULTS

1. Experiments Regarding 30 Days of Immobilization

The average values of tension parameters obtained from Sartorius muscles (Table 1 and Fig. 1) showed significantly ($P < 0.005$) reduced twitch (52%) and tetanic (43%) tensions in immobilized muscles as compared with their respective controls. Similarly, twitch/tetanus ratio was also reduced (14%) significantly ($P < 0.01$) after immobilization (Table 1 and Fig. 1).

Twitch contraction time (TwCT) was also reduced (15%) significantly ($P < 0.05$) after 30 days of immobilization. On the other hand, the duration of twitch peak (TwPD) and active state (DAS) showed insignificant reduction by about 10%, after immobilization (Table 1 and Fig. 2).

2. Experiments Regarding Remobilization

At the end of 30 days of immobilization the castings were removed and animals were allowed to remobilize for 10 days. The average values of tension parameters obtained from control and remobilized Sartorius muscles are presented in Table 1 and Fig. 1. These values demonstrated that after 10 days of remobilization twitch and tetanic tensions as well as their ratio were only 4-12% less in remobilized muscles than their respective controls. But, this difference between them was statistically insignificant (Table 1 and Fig. 1).

Table 1 Average values of tension and time parameters measured from twitch and tetanus records obtained from control and immobilized/remobilized muscles of *Uromastix hardwickii*

S. Parameters No.	Control	Immobilized	P*	Control	Remobilized	P*
1. Twitch Tension	0.489±0.038 (11)	0.233±0.019 (10)	P<0.005	0.605±0.049 (11)	0.532±0.031 (11)	P>0.05
2. Tetanic Tension	1.214±0.081 (12)	0.685±0.035 (10)	P<0.005	1.356±0.102 (11)	1.301±0.09 (11)	P>0.05
3. Twitch/Tetanus Ratio	0.392±0.015 (11)	0.337±0.015 (10)	P<0.01	0.448±0.019 (11)	0.415±0.017 (11)	P>0.05
4. Contraction Time	0.065±0.004 (12)	0.055±0.003 (12)	P<0.05	0.066±0.003 (11)	0.059±0.002 (11)	P<0.05
5. Peak Duration	0.017±0.001 (12)	0.015±0.001 (12)	P>0.05	0.014±0.001 (11)	0.014±0.001 (11)	P>0.05
6. Active State Duration	0.028±0.001 (12)	0.025±0.002 (12)	P>0.05	0.025±0.001 (11)	0.026±0.002 (11)	P>0.05

P*: Significance levels between control and immobilized/remobilized muscles.
Figures in the parenthesis represents number of muscles used.

EFFECT OF IMMOBILIZATION AND REMOBILIZATION ON TENSION PARAMETERS

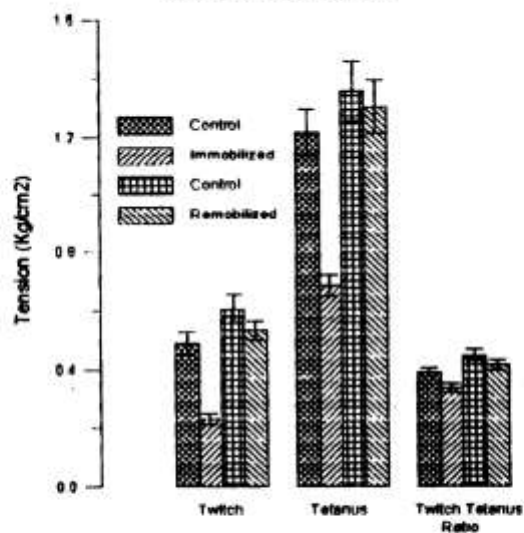


Fig 1

EFFECT OF IMMOBILIZATION AND REMOBILIZATION ON TIME PARAMETERS

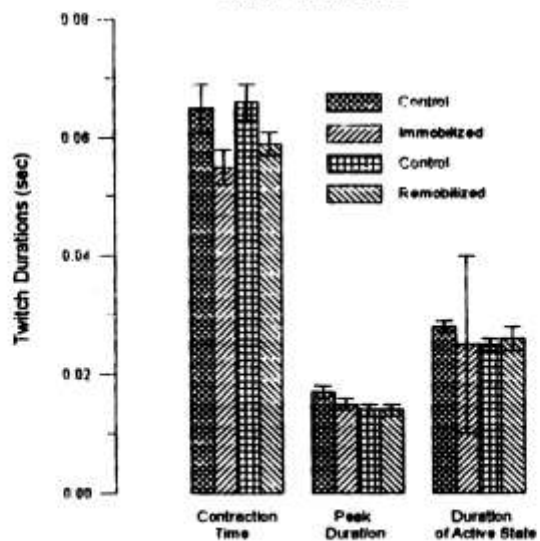


Fig 2

TwCT demonstrated significant ($P < 0.05$) reduction (11%) than its respective control, after 10 days of remobilization. The other two parameters i.e., TwPD and DAS did not showed any significant change (Table and Fig. 2) when compared between remobilized and their respective controls.

DISCUSSION

A number of previous studies revealed that limb immobilization results in muscular atrophy (Edgerton *et al.*, 1975; Maxwell and Enwemwka, 1992) related with reduced contraction parameters. Our experimental results showed that 30 days of immobilization in Sartorius muscles (neutral position) was unable to produce muscular atrophy. This observation is based on insignificant changes in muscle weight and also in agreement with a previous study reported by Azeem *et al.*, 1993. They also demonstrated insignificant atrophy in Gastrocnemius muscles of *Uromastix hardwickii* after short term hind limb immobilization.

Position of muscle during immobilization greatly affects the degree of atrophy (Spector *et al.*, 1982; Syed *et al.*, 1991). Spector *et al.*, (1982) reported that immobilization in neutral position causes greater loss of muscle weight than that in lengthened position. In the present study Sartorius muscle was also immobilized in neutral position, but instead weight loss a significant increase in muscle weight was observed. On the basis of this observation and also on the previous study reported by Azeem *et al.*, (1993), we suggested that skeletal muscles of *Uromastix hardwickii* are resistant to immobilization atrophy, and capable to increase their weight, dependent on their position during immobilization.

Since, we found no significant change in muscles's stretched and flexed lengths following immobilization therefore a possibility of sarcomere addition (Tabary *et al.*, 1972) is out of question. Hence increase in muscle weight after immobilization probably represents synthesis of non-contractile proteins. This suggestion is based on the report of Karpakka *et al.*, (1991) that one week of immobilization in Tibialis Anterior muscles of rat results in an increased collagen contents in experimental muscles.

Moreover, twitch and tetanic tensions as well as their ratio reduced significantly after immobilization (30 days). This decline in tension production ability is analogous to those reported in previous studies (Simard *et al.*, 1982; Witzmann *et al.*, 1982; Cotter *et al.*, 1991; and Azeem *et al.*, 1993). In our opinion there could be other two possibilities for this reduction in tension parameters after immobilization in Sartorius muscles of *Uromastix*:

1. reduction in contractile to non-contractile protein ration
2. reduction in concentration or availability of various enzymes involved in muscle mechanics.

Witzmann *et al.*, (1982) however, attributed this decline in tension to a reduction in the number of active cross-bridges after immobilization in rat hind limb muscles. On the other hand Azeem *et al.*, (1993) related it with reduction in energy producing substrates and sarcoplasmic capacity for calcium manipulation in Gastrocnemius muscles of *Uromastix hardwickii* after short term immobilization. Regarding the first possibility a probable increase in non-contractile proteins in our experimental muscles resulted in reduction in their tensions when calculated in units of kg/cm² (Table 1). Further, increased non-contractile components in the form of collagen (Karpakka *et al.*, 1991) raised muscle stiffness that hindered tension production during cross-bridge interaction. Sarcoplasmic concentration is also increased following immobilization (Diffie *et al.*, 1991).

Booth and Seider (1979) reported recovery in biochemical parameters in immobilized rat Soleus muscles. Witzmann *et al.*, (1982) demonstrated complete recovery in various contractile functions of rat muscles. Our experimental muscles which were allowed to recover for 10 days after immobilization demonstrated insignificantly different tension parameters from their respective controls. Thus, representing recovery of about 75% to 90% in tension and 50% in their ratio (Table 1). This recovery in tension parameters should not be due to alteration in contractile or myosin ATPase activity since, during immobilization tension was reduced (suggested decreased contractile to non-contractile components ratio) while, muscle weight was higher (suggested increased non-contractile components) with respect to controls. Therefore, the only hypothetical possibility regarding decrease in muscle weight along with increased tensions is the degradation of non-contractile components (of collagen by collagenases). However, this hypothesis remains to be tested.

The TwCT recovered by 27% only and remained significantly less than control as was observed during immobilization. While TwPD and DAS showed 100% recovery on remobilization, compared to their respective parameters obtained during immobilization (Table 1). However, in the case of DAS the recovery followed 4% rise in its average value (in remobilized muscles) than its control, which was previously reduced during immobilization (Table 1). According to above discussion our experimental muscles demonstrated earlier recovery intension parameters than twitch contraction time which is in contrast to the reports of Witzmann *et al.*, (1982) who demonstrated earlier recovery in time parameters as compared to tetanic tension.

It is concluded that skeletal muscles of *Uromastix hardwickii* are resistant to immobilization atrophy. Impaired mechanical activity probably showed alteration in contractile to non-contractile protein ratio and lesser ATPase activity in immobilized muscles. The tension parameters recovered earlier than twitch contraction time. The histological and biochemical studies are suggested to strengthen this conclusion.

REFERENCES

- Arifa, S., Azeem, M.A., Shahina, S., Shaikh, K.N. and Shaikh, H.A., (1993). Mechanical response of lizard skeletal muscles to disuse: 1. Effect of short-term tenotomization, *Acta Physiologica Hungarica*, **81**(3): 301-307.
- Azeem, M.A., Irfan, T., Fehmeena, S. Arifa, S. Najamuddin, F. and Shaikh, H.A., (1993). Effect of short term immobilization on mechanical characteristics of skeletal muscle in *Uromastix*, *Pak. J. Pharma. Sci.*, **6**(2): 9-17.
- Azeem, M.A. and Shaikh, H.A., (1987). Correlation between the anatomical and mechanical characteristics of gastrocnemius and sartorius muscles of *Uromastix hardwickii*, *Acta Biologica Cracoviensis*, **29**: 105-114.
- Booth, F.W. and Kelso, J.R., (1973). Effect of hind limb immobilization on contractile and histochemical properties of skeletal muscle, *Pfluegers Arch Eur. J. Physiol.*, **342**(3): 231-238.
- Booth, F.W. and Seider, M.J., (1979). Recovery of skeletal muscle after 3 months of hind limb immobilization in rats, *J. Appl. Physiol.; Respirat. Environ. Exercise Physiol.*, **47**(2): 435-439.
- Cotter, M.A. Cameron, N.E. Barry, J.A. and Pattullo, M.C., (1991). Chronic stimulation accelerates functional recovery of immobilized soleus muscles of the rabbit, *Expt. Physiol.*, **76**(2): 201-212.
- Diffie, G.M., Caiozzo, V.J., Herrick, R.E. and Baldwin, K.M., (1991). Contractile and biochemical properties of rat soleus and plantaris after hind limb suspension, *Am. J. Physiol.*, **260**(3): C528-C534.
- Edgerton, V.R., Barnard, R.J., Peter, J.B., Maier, A. and Simpson, D.R., (1975). Properties of immobilized hind limb muscles of the Galago Senegalensis, *Exp. Neurol.*, **46**: 115-131.
- Finkelstein, D.I., Dooley, P.C. and Luff, A.R., (1993). Recovery of muscle after different periods of denervation and treatments, *Muscle and Nerve*, **16**: 769-777.
- Kannus, P., Jozsa, L., Renstrom, P., Jarvinen, M., Kvist, M., Lehto, M., Oja, P. and Vuori, I., (1992). The effects of training, immobilization and remobilization on musculoskeletal tissue: 1. Training and immobilization, *Scand. J. Med. Sci. Sports.*, **2**(3): 100-118.
- Karpakka, J., Virtanen, P., Vaananen, K., Orava, S. and Takala, T.E.S. (1991). Collagen synthesis in rat skeletal muscle during immobilization and remobilization. *J. Appl. Physiol.*, **70**(4): 1775-1780.
- Lindboe, C.F. and Presthus, J., (1985). Effects of denervation, immobilization and cachexia on fibre size in anterior tibial muscle of the rat, *Acta Neuropathol.*, **66**(1): 42-51.
- Maxwell, L.C. and Enwemeka, C.S., (1992). Immobilization-induced muscle atrophy is not reversed by lengthening the muscle, *Anat. Rec.*, **234**(1): 55-61.

REFERENCES

- Arifa, S., Azeem, M.A., Shahina, S., Shaikh, K.N. and Shaikh, H.A., (1993). Mechanical response of lizard skeletal muscles to disuse: 1. Effect of short-term tenotomization, *Acta Physiologica Hungarica*, **81**(3): 301-307.
- Azeem, M.A., Irfan, T., Fehmeena, S. Arifa, S. Najamuddin, F. and Shaikh, H.A., (1993). Effect of short term immobilization on mechanical characteristics of skeletal muscle in *Uromastix*, *Pak. J. Pharma. Sci.*, **6**(2): 9-17.
- Azeem, M.A. and Shaikh, H.A., (1987). Correlation between the anatomical and mechanical characteristics of gastrocnemius and sartorius muscles of *Uromastix hardiwickii*, *Acta Biologica Cracoviensis*, **29**: 105-114.
- Booth, F.W. and Kelso, J.R., (1973). Effect of hind limb immobilization on contractile and histochemical properties of skeletal muscle, *Pfluegers Arch Eur. J. Physiol.*, **342**(3): 231-238.
- Booth, F.W. and Seider, M.J., (1979). Recovery of skeletal muscle after 3 months of hind limb immobilization in rats, *J. Appl. Physiol.; Respirat. Environ. Exercise Physiol.*, **47**(2): 435-439.
- Cotter, M.A. Cameron, N.E. Barry, J.A. and Pattullo, M.C., (1991). Chronic stimulation accelerates functional recovery of immobilized soleus muscles of the rabbit, *Expt. Physiol.*, **76**(2): 201-212.
- Diffie, G.M., Caiozzo, V.J., Herrick, R.E. and Baldwin, K.M., (1991). Contractile and biochemical properties of rat soleus and plantaris after hind limb suspension, *Am. J. Physiol.*, **260**(3): C528-C534.
- Edgerton, V.R., Barnard, R.J., Peter, J.B., Maier, A. and Simpson, D.R., (1975). Properties of immobilized hind limb muscles of the Galago Senegalensis, *Exp. Neurol.*, **46**: 115-131.
- Finkelstein, D.I., Dooley, P.C. and Luff, A.R., (1993). Recovery of muscle after different periods of denervation and treatments, *Muscle and Nerve*, **16**: 769-777.
- Kannus, P., Jozsa, L., Renstrom, P., Jarvinen, M., Kvist, M., Lehto, M., Oja, P. and Vuori, I., (1992). The effects of training, immobilization and remobilization on musculoskeletal tissue: 1. Training and immobilization, *Scand. J. Med. Sci. Sports.*, **2**(3): 100-118.
- Karpakka, J., Virtanen, P., Vaananen, K., Orava, S. and Takala, T.E.S. (1991). Collagen synthesis in rat skeletal muscle during immobilization and remobilization. *J. Appl. Physiol.*, **70**(4): 1775-1780.
- Lindboe, C.F. and Presthus, J., (1985). Effects of denervation, immobilization and cachexia on fibre size in anterior tibial muscle of the rat, *Acta Neuropathol.*, **66**(1): 42-51.
- Maxwell, L.C. and Enwemeka, C.S., (1992). Immobilization-induced muscle atrophy is not reversed by lengthening the muscle, *Anat. Rec.*, **234**(1): 55-61.